Full CMOS Implementation of Bidirectional Associative Memory Neural Network with Analog Memristive Synapse

B.Tech Thesis Presentation

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Outline

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What is Neuromorphic Computing & Why Use It?

The Von Neumann Bottleneck

- Conventional computing separates CPU and Memory.
- Constant data movement causes high latency and massive energy consumption.
- Modern AI requires Kilowatts of power.

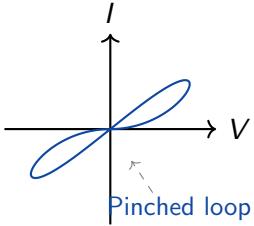
The Neuromorphic Solution

- **Brain-Inspired:** Brain processes and stores data in synapses using only $\sim$20W.
- **Neuromorphic Computing:** Hardware mimics biological neural networks.
- Highly parallel, event-driven, integrates memory with computation.

What is a Memristor?

The Fourth Fundamental Element

- Postulated by L.O. Chua (1971).
- Relates **magnetic flux** (φ) and **charge** (q).
- Resistance depends on history of current — non-volatile memory.



Key Property

$M = \frac{d\varphi}{dq} \quad \Rightarrow \quad$ Memristance

Pinched hysteresis loop in I-V characteristic

The 4 elements:

R (V-I) L (Flux-I) C (Charge-V) M (Flux-Charge)

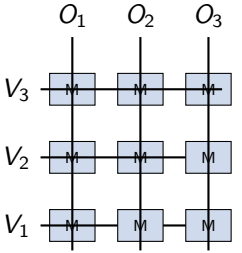
Memristors as Artificial Synapses

Why Memristors?

- Analog tunable resistance → **synaptic weight**
- Non-volatile → **memory without power**
- High density crossbar arrays
- In-memory computation

Hardware Challenge

Physical memristors are **not available** in standard CMOS PDKs due to fabrication complexities → Need a CMOS emulator!



Memristive Crossbar

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Literature Review & Motivation

Work	Approach	Limitation
Zhao et al. (ISPACS 2017)	Memristor SPICE model	Hardware not fabricatable
Shi & Zeng (IEEE 2020)	In-situ learning BAM	SPICE model only
Saxena (MWSCAS 2019)	Compact CMOS emulator	Applied to memcomputing
This Work	**Full CMOS BAM, 180nm**	**First full CMOS implementation**

Research Gap & Motivation

- No prior work implemented a **complete CMOS circuit** of BAM using a CMOS memristor emulator in standard technology.
- Physical stochastic device characteristics limit reliable weight storage.

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Problem Statement & Objectives

Problem Statement

To design and simulate a **6×6 Bidirectional Associative Memory** neural network using a **CMOS-based tunable memristor emulator** as the synaptic element in **SCL 180nm CMOS technology**.

Specific Objectives

1. **Objective 1:** Implement a CMOS memristor emulator with tunable resistance to act as an artificial synapse.
2. **Objective 2:** Build a fully analog 6×6 BAM crossbar synapse circuit.
3. **Objective 3:** Resolve analog TIA loading issues for stable logic outputs.
4. **Objective 4:** Demonstrate successful pattern learning and bidirectional recall of spatial patterns via Virtuoso simulations.

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Proposed Technique: BAM Algorithm & Weight Calculation

Two-Layer Neural Network

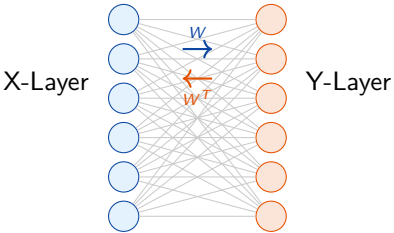
- X-layer (n neurons) $\leftrightarrow$ Y-layer (p neurons)
- Fully connected via weight matrix W
- Bidirectional: input from either layer

$$y_{\text{next}} = f(xW, x), \quad x_{\text{next}} = f(yW^T, y)$$

Hebbian Learning Rule

$$W = \sum_{k=1}^{M} X_k^T Y_k$$

Logical weights: $W_{jk} \in \{-3, -1, +1, +3\}$



Tetris pattern pairs used:
Black box $= +1$, White box $= -1$

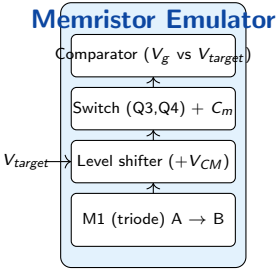
Implementation: CMOS Memristor Emulator

Tunable Resistance Mechanism

Transistor M1 in deep triode region:

$$G = K_n \frac{W}{L}(V_{GS} - V_{CM} - V_{Tn})$$

- W/L = 0.42 μm / 10 μm
- VGS controls resistance continuously



Programming Circuit

1. Gm block senses V_{AB}, charges C_m
2. Comparator monitors V_g vs V_{target}
3. Switch opens when $V_g = V_{target}$
4. Level shifter adds V_{CM} offset

Implementation: Crossbar Architecture & Mapping

1M Crossbar Synapse

Output at Y-layer column k:

$$V_{ok} = \sum_{j=1}^{p} R_0(G_s - G_{jk})V_{ij}$$

Weight to Resistance Mapping

Weight	Resistance	V_{target}
$+1$	95.2 kΩ	700 mV
-1	84 kΩ	830 mV
$+3$	110 kΩ	584 mV
-3	75 kΩ	970 mV

6×6 Weight Matrix Generated

$$W = \begin{bmatrix} -1 & -1 & -1 & +1 & +1 & +3 \\ +1 & +1 & -3 & +3 & +3 & +1 \\ +1 & +1 & -3 & +3 & +3 & +1 \\ -1 & -1 & +3 & -3 & -3 & -1 \\ -1 & +3 & -1 & +1 & +1 & -1 \\ +3 & -1 & -1 & +1 & +1 & -1 \end{bmatrix}$$

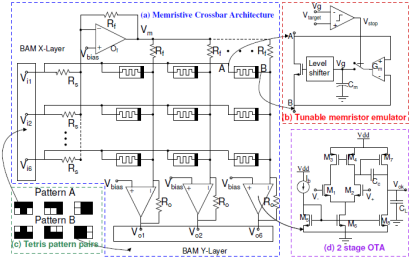
Two-Stage OTA

Used for: (a) virtual ground Vm, (b) 6 column TIAs

PM2 width = 4 μm for balanced 2nd stage

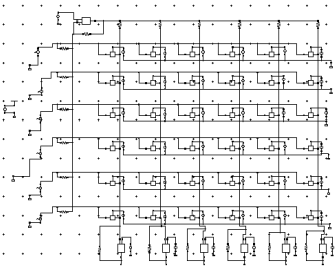
Circuit Implementation: Reference vs. Our Design

Reference Architecture



Proposed theoretical BAM

Our 180nm CMOS Implementation



Full Analog Virtuoso Schematic

- **Transition to Silicon:** We realized the theoretical 1M Crossbar by modeling physical PWL sources, two-stage OTAs, and tuning transimpedance parameters.

Results: Learning Phase (Memristor Programming)

Training Protocol (0 – 5 μs)

- V_{target} PWL sources pulse each cell
- Comparator charges C_m toward V_{target}
- Switch opens when $V_g = V_{target}$
- Weight locked non-volatilely on C_m

Capacitor Voltage Behavior

- 0–5 μs: C_m charges to V_{target}
- 5.001 μs: V_{target} drops to 0 V (locked)

Programming Time

Target resistance reached rapidly.
5 μs training window provides ample time to securely lock weights. ✓

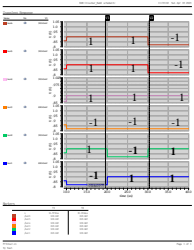
Circuit Design Challenges Solved

1. **OTA imbalance**: 2nd stage PMOS width increased to 4μm.
2. **Virtual ground**: Separate Vm summing node implements crossbar properly.
3. **Switch topology**: Complementary pass-gates ensure full weight swing.

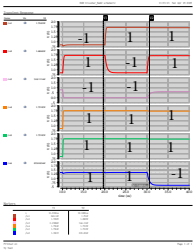
Results: Simulation Waveforms (Pattern Recall)

Transient Response (10 μs – 40 μs Inference Window)

Input Pattern A: $[+1, +1, +1, -1, +1, -1] \rightarrow$ Expected B: $[-1, +1, -1, +1, +1, +1]$



Input Pattern Pulses



Output Recalled Pattern

- **Activation:** Outputs $> 1\,\mathrm{V}$ evaluate to logic $+1$, outputs $< 1\,\mathrm{V}$ to logic -1.
- The graph successfully demonstrates the expected logic exact pattern matches.

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Conclusion & Future Work

Conclusions

- Successfully implemented **6×6 BAM** in standard 180 nm CMOS.
- CMOS memristor provides **tunable non-volatile weight storage**.
- Full simulation demonstrates **correct pattern association**.
- **First full CMOS implementation** of BAM with a CMOS synapse.

Future Work

- **Layout design** and post-layout simulation.
- Extend to **larger BAM networks** (12×12, 24×24) for image recognition.
- Implement **on-chip STDP learning** for autonomous updates.
- Power analysis and **energy optimization**.

References

- L. Chua, "Memristor-the missing circuit element," *IEEE Trans. Circuit Theory*, vol. 18, no. 5, pp. 507–519, 1971.
- R. S. Williams, "How We Found The Missing Memristor," *IEEE Spectrum*, vol. 45, no. 12, pp. 28–35, 2008.
- S. K. Vohra, S. Thomas, M. Sakare, D. M. Das, "Full CMOS Implementation of Bidirectional Associative Memory Neural Network with Analog Memristive Synapse," *IEEE MWSCAS*, pp. 445–448, 2021.
- S. N. Truong, K. Min, "New memristor-based crossbar array architecture with 50% area reduction," *J. Semiconductor Technol. Sci.*, vol. 14, no. 3, pp. 356–363, 2014.
- V. Saxena, "A compact CMOS memristor emulator circuit and its applications," *IEEE MWSCAS*, pp. 190–193, 2019.

Thank You

Questions & Discussion

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